

- If no first-line atypical antipsychotic is effective, consider higher doses or augmentation with valproate or lamotrigine
- Consider noncompliance and switch to another antipsychotic with fewer side effects or to an antipsychotic that can be given by depot injection
- Consider initiating rehabilitation and psychotherapy such as cognitive remediation
- Consider presence of concomitant drug abuse

Best Augmenting Combos for Partial Response or Treatment Resistance

- Valproic acid (valproate, divalproex, divalproex ER)
- Other mood-stabilizing anticonvulsants (carbamazepine, oxcarbazepine, lamotrigine)
- Topiramate
- Lithium
- Benzodiazepines

Tests

- Baseline ECG, serum potassium, and serum magnesium measurements
- ECG should be repeated upon reaching steady state about 3 weeks after treatment initiation or upon reaching 16 mg/day, at 3 months,

and then every 3 months for the duration of treatment

- ECG is recommended prior to and after any dose increase as well as after addition of any drug that can change sertindole concentration (e.g., CYP450 2D6 or 3A4 inhibitors)

Before starting an atypical antipsychotic

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient
 - is overweight (BMI 25.0–29.9)
 - is obese (BMI >30)
 - has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
 - has diabetes (fasting plasma glucose >126 mg/dL)
 - has hypertension (BP >140/90 mmHg)
 - has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting an atypical antipsychotic

- * BMI monthly for 3 months, then quarterly
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight
- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic
- * Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness, and clouding of sensorium, even coma
- Should check blood pressure during titration and early maintenance treatment
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

- * Orthostatic hypotension
- Dizziness, dry mouth, nasal congestion
- Weight gain, peripheral edema, decreased ejaculatory volume
- * May increase risk for diabetes and dyslipidemia
- Tardive dyskinesia (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration



Life-Threatening or Dangerous Side Effects

- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis
- Rare seizures

Weight Gain



- Many experience and/or can be significant in amount
- May be less than for some antipsychotics, more than for others

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait

- Wait
- Wait
- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 12–20 mg/day

Dosage Forms

- Tablet 4 mg, 12 mg, 16 mg, 20 mg

How to Dose

- Initial 4 mg/day; increase by 4 mg every 4–5 days; maximum dose generally 24 mg/day

Dosing Tips

- A starting dose of 8 mg or a rapid increase in dose significantly increase risk of orthostatic hypotension
- Sertindole should be discontinued if QTc interval of more than 500 msec is observed during treatment

Overdose

- Fatalities have been reported; sedation, slurred speech, tachycardia, hypotension, transient QTc prolongation; torsade de pointes

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Down-titration, especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid discontinuation could theoretically lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Terminal half-life approximately 3 days
- Extensively metabolized by CYP450 2D6 and 3A



Drug Interactions

- May increase effects of antihypertensive agents
- May antagonize levodopa, dopamine agonists
- May enhance QTc prolongation of other drugs capable of prolonging QTc interval
- Inhibitors of CYP450 2D6 (e.g., paroxetine, fluoxetine, duloxetine, quinidine) may significantly increase plasma levels of sertindole and require a dosage reduction of sertindole
- Inhibitors of CYP450 3A (e.g., nefazodone, fluvoxamine, fluoxetine, ketoconazole) can increase plasma levels of sertindole and concomitant use of these agents with sertindole is contraindicated

Other Warnings/Precautions

- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and sertindole should be used cautiously in patients at risk for aspiration pneumonia
- Sertindole prolongs QTc interval more than some other antipsychotics
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is taking agents capable of significantly prolonging QTc interval (e.g., pimozide, thioridazine, selected antiarrhythmics, moxifloxacin, sparfloxacin)
- If there is a personal or family history of QTc prolongation
- If there is a history of significant cardiovascular illness, including congestive heart failure, cardiac hypertrophy, arrhythmia, bradycardia, or congenital prolonged QT syndrome
- If patient has hypokalemia, hypomagnesemia, or QTc greater than 450 msec (males) or 470 msec (females)
- If patient is taking a CYP450 3A inhibitor
- If patient has severe hepatic impairment
- If patient has galactose intolerance, Lapp lactase deficiency, or glucose-galactose malabsorption
- If there is a proven allergy to sertindole

Special Populations

Renal Impairment

- Dose adjustment not generally necessary

Hepatic Impairment

- For mild to moderate impairment, use slower titration and lower maintenance dose

- Contraindicated in patients with severe hepatic impairment

Cardiac Impairment

- Drug should be used with caution because of risk of orthostatic hypotension
- Not recommended for patients with significant cardiovascular illness, including congestive heart failure, cardiac hypertrophy, arrhythmia, bradycardia, or congenital prolonged QT syndrome

Elderly

- Use cautiously and only after thorough cardiovascular examination
- Some patients may tolerate lower doses better
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events



Children and Adolescents

- Safety and efficacy have not been established
- Children and adolescents using sertindole may need to be monitored more often than adults and may tolerate lower doses better

Pregnancy

- Some animal studies show adverse effects; no controlled studies in humans
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary

Breast Feeding

- Unknown if sertindole is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Infants of women who choose to breast feed while on sertindole should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- Some cases of psychosis and bipolar disorder refractory to treatment with other antipsychotics

Potential Disadvantages

- Patients requiring rapid onset of antipsychotic action without dosage titration

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive symptoms
- Unstable mood (both depression and mania)
- Aggressive symptoms

Pearls

- Aimed at patients intolerant to at least one other antipsychotic and when potential benefits outweigh potential risks
- Not approved for mania, but all atypical antipsychotics approved for acute schizophrenia are generally also useful for acute bipolar mania
- Not approved for depression, but binding properties suggest potential use in treatment-resistant and bipolar depression, though probably not as a first- or second-line agent given the need to accumulate more real-world clinical experience of any actual risks of sertindole

Suggested Reading

Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

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Komossa K , Rummel-Kluge C , Hunger H , et al. Sertindole versus other atypical antipsychotics for schizophrenia . *Cochrane Database Syst Rev* 2009 ;(**2**) :CD006752 .

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Lindstrom E , Farde L , Eberhard L , Haverkamp W . QTc interval prolongation and antipsychotic drug treatments: focus on sertindole . *Int J Neuropsychopharmacol* 2005 ;**8** :615–29 .

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Sertraline

Therapeutics

Brands

- Zoloft
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: serotonin reuptake inhibitor (S-RI)
- SSRI (selective serotonin reuptake inhibitor); often classified as an antidepressant, but it is not just an antidepressant

Commonly Prescribed for

(bold for FDA approved)

- **Major depressive disorder**
- **Premenstrual dysphoric disorder (PMDD)**
- **Panic disorder**
- **Posttraumatic stress disorder (PTSD)**

- **Social anxiety disorder (social phobia)**
- **Obsessive-compulsive disorder (OCD)**
- Generalized anxiety disorder (GAD)

How the Drug Works

- Boosts neurotransmitter serotonin
- Blocks serotonin reuptake pump (serotonin transporter)
- Desensitizes serotonin receptors, especially serotonin 1A receptors
- Presumably increases serotonergic neurotransmission
- * Sertraline also has some ability to block dopamine reuptake pump (dopamine transporter), which could increase dopamine neurotransmission and contribute to its therapeutic actions
- Sertraline also binds at sigma 1 receptors

How Long Until It Works

- * Some patients may experience increased energy or activation early after initiation of treatment
- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6–8 weeks, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission) or significantly reduced (e.g., OCD, PTSD)
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders may also need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating in depression)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent

- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- Trazodone, especially for insomnia
- In the USA, sertraline (Zoloft) is commonly augmented with bupropion (Wellbutrin) with good results in a combination anecdotally called “Well-loft” (use combinations of antidepressants with caution as this may activate bipolar disorder and suicidal ideation)
- Mirtazapine, reboxetine, or atomoxetine (add with caution and at lower doses since sertraline could theoretically raise atomoxetine levels); use combinations of antidepressants with caution as this may activate bipolar disorder and suicidal ideation
- Modafinil, especially for fatigue, sleepiness, and lack of concentration
- Mood stabilizers or atypical antipsychotics for bipolar depression, psychotic depression, treatment-resistant depression, or treatment-resistant anxiety disorders

- Benzodiazepines
- If all else fails for anxiety disorders, consider gabapentin or tiagabine
- Hypnotics for insomnia
- Classically, lithium, buspirone, or thyroid hormone

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Theoretically due to increases in serotonin concentrations at serotonin receptors in parts of the brain and body other than those that cause therapeutic actions (e.g., unwanted actions of serotonin in sleep centers causing insomnia, unwanted actions of serotonin in the gut causing diarrhea, etc.)
- * Increasing serotonin can cause diminished dopamine release and might contribute to emotional flattening, cognitive slowing, and apathy in some patients, although this could theoretically be diminished in some patients by sertraline's dopamine reuptake blocking properties
- Most side effects are immediate but often go away with time, in contrast to most therapeutic effects which are delayed and are enhanced over time

- Sertraline's possible dopamine reuptake blocking properties could contribute to agitation, anxiety, and undesirable activation, especially early in dosing

Notable Side Effects

- Sexual dysfunction (dose-dependent; men: delayed ejaculation, erectile dysfunction; men and women: decreased sexual desire, anorgasmia)
- Gastrointestinal (decreased appetite, nausea, diarrhea, constipation, dry mouth)
- Mostly CNS (insomnia but also sedation, agitation, tremors, headache, dizziness)
- Note: patients with diagnosed or undiagnosed bipolar or psychotic disorders may be more vulnerable to CNS-activating actions of SSRIs
- Autonomic (sweating)
- Bruising and rare bleeding
- Rare hyponatremia (mostly in elderly patients and generally reversible on discontinuation of sertraline)
- Rare hypotension
- SIADH (syndrome of inappropriate antidiuretic hormone secretion)



Life-Threatening or Dangerous Side Effects

- Rare seizures
- Rare induction of mania
- Rare activation of suicidal ideation and behavior (suicidality) (short-term studies did not show an increase in the risk of suicidality with antidepressants compared to placebo beyond age 24)

Weight Gain



- Reported but not expected
- Some patients may actually experience weight loss

Sedation



- Reported but not expected
- Possibly activating in some patients

What to Do About Side Effects

- Wait
- Wait
- Wait
- If sertraline is activating, take in the morning to help reduce insomnia

- Reduce dose to 25 mg or even 12.5 mg until side effects abate, then increase dose as tolerated, usually to at least 50 mg/day
- In a few weeks, switch or add other drugs

Best Augmenting Agents for Side Effects

- Often best to try another SSRI or another antidepressant monotherapy prior to resorting to augmentation strategies to treat side effects
- Trazodone or a hypnotic for insomnia
- Bupropion, sildenafil, vardenafil, or tadalafil for sexual dysfunction
- Bupropion for emotional flattening, cognitive slowing, or apathy
- Mirtazapine for insomnia, agitation, and gastrointestinal side effects
- Benzodiazepines for jitteriness and anxiety, especially at initiation of treatment and especially for anxious patients
- Many side effects are dose-dependent (i.e., they increase as dose increases, or they reemerge until tolerance redevelops)
- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of sertraline

Dosing and Use

Usual Dosage Range

- 50–200 mg/day

Dosage Forms

- Tablet 25 mg scored, 50 mg scored, 100 mg
- Oral solution 20 mg/mL

How to Dose

- Depression and OCD: initial 50 mg/day; usually wait a few weeks to assess drug effects before increasing dose, but can increase once a week by 50 mg/day; maximum generally 200 mg/day; single dose
- Panic, PTSD, and social anxiety: initial 25 mg/day; increase to 50 mg/day after 1 week thereafter, usually wait a few weeks to assess drug effects before increasing dose, but can increase once a week by 25–50 mg/day; maximum generally 200 mg/day in a single dose
- PMDD: initial 50 mg/day; can dose daily through the menstrual cycle (up to 150 mg/day) or limit to the luteal phase (up to 100 mg/day)
- Oral solution: mix with 4 oz of water, ginger ale, lemon/lime soda, lemonade, or orange juice only; drink immediately after mixing



Dosing Tips

- All tablets are scored, so to save costs, give 50 mg as half of 100 mg tablet, since 100 mg and 50 mg tablets cost about the same in many markets
- Give once daily, often in the mornings to reduce chances of insomnia
- Many patients ultimately require more than 50 mg dose per day
- Some patients are dosed above 200 mg
- Evidence that some treatment-resistant OCD patients may respond safely to doses up to 400 mg/day, but this is for experts and use with caution
- The more anxious and agitated the patient, the lower the starting dose, the slower the titration, and the more likely the need for a concomitant agent such as trazodone or a benzodiazepine
- If intolerable anxiety, insomnia, agitation, akathisia, or activation occur upon either dosing initiation or discontinuation, consider the possibility of activated bipolar disorder and switch to a mood stabilizer or atypical antipsychotic
- Utilize half a 25 mg tablet (12.5 mg) when initiating treatment in patients with a history of intolerance to previous antidepressants

Overdose

- Rarely lethal in monotherapy overdose; vomiting, sedation, heart rhythm disturbances, dilated pupils, agitation; fatalities have been reported in sertraline overdose combined with other drugs or alcohol

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper to avoid withdrawal effects (dizziness, nausea, stomach cramps, sweating, tingling, dysesthesias)
- Many patients tolerate 50% dose reduction for 3 days, then another 50% reduction for 3 days, then discontinuation
- If withdrawal symptoms emerge during discontinuation, raise dose to stop symptoms and then restart withdrawal much more slowly

Pharmacokinetics

- Parent drug has 22–36 hour half-life
- Metabolite half-life 62–104 hours
- Inhibits CYP450 2D6 (weakly at low doses)
- Inhibits CYP450 3A4 (weakly at low doses)



Drug Interactions

- Tramadol increases the risk of seizures in patients taking an antidepressant